2 type of loss

Unit-3

Single Phase Transformer

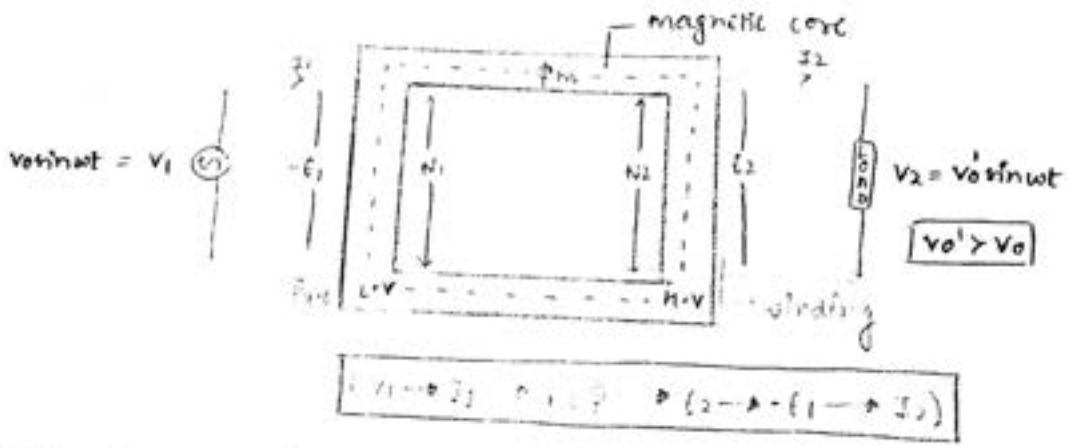
Transformer :-

It is an static device which is used to transform an electric energy from one circuit to another circuit by increase or decrease in voltage according to the principle of mutual induction without changing its frequency.

Transformer is of two types.

1. Step up transformer (increase in magnitude of voltage)
2. Step down transformer (decrease in magnitude of voltage)

Working Principle of Transformer :-



When AC supply is given to primary coil then current will be produced due to which flux will be created in primary coil according to Lenz's law the flux starared to moves in upward direction towards secondary coil we know, when flux is goes to secondary coil. Emf (E_2) is produced in that coil due to Electromagnetic induction law or Faraday's law. Similarly ($-E_1$) Emf is produced in primary coil.

The negative sign in ($-E_1$) shows that the produced Emf opposed the applied Emf/voltage.

When load is connected to the secondary coil the current (I_2) start moving towards load.

Q. Why transformer is never operated with DC supply?

A. The AC flux is produced in the primary coil due to primary current in the coil. When AC flows in primary coil, current is induced in the secondary coil by the law of mutual induction. There is no mutual induction in DC as the direction of current or flux flow does not change. So a transformer can not work in DC.

In primary coil there is both resistance as well as inductor. When we give DC supply then frequency is equals to zero

We know,

$$X_L = \omega L$$
$$= 2\pi f L = 2\pi(0)L$$
$$\boxed{X_L = 0}$$

So primary coil starts behaving as a pure resistance, and we know that the resistance in the coil is very low. So the minimum resistance is equals to maximum current.

Due to max current flow, in primary coil current will not be able to flow and will burn out.

Emf equation of Transformer :-

According to Faraday's law

$$E \propto -N\frac{d\phi}{dt}$$

$$\phi = \phi_0 \sin\omega t$$

$$E \propto -N\frac{d\phi}{dt}\sin\omega t$$

$$E \propto -N\phi_0\omega\cos\omega t$$

$$E \propto -N\phi_0\omega(\sin\omega t - \pi/2)$$

We know,

$$E = E_0 \sin\omega t \pm \phi$$

On comparision

$$E_0 = N\phi_0 2\pi f$$

$$E_{rms} = \frac{E_0}{\sqrt{2}} = \frac{N\phi_0 2\pi f}{\sqrt{2}}$$

$$\boxed{E_{rms} = 4.44 N\phi_0 f}$$

$$\frac{E_2}{E_1} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

K = Transforming ratio.

$$\boxed{E_1 = 4.44 N_1\phi_0 f}$$

$$\boxed{E_2 = 4.44 N_2\phi_0 f}$$

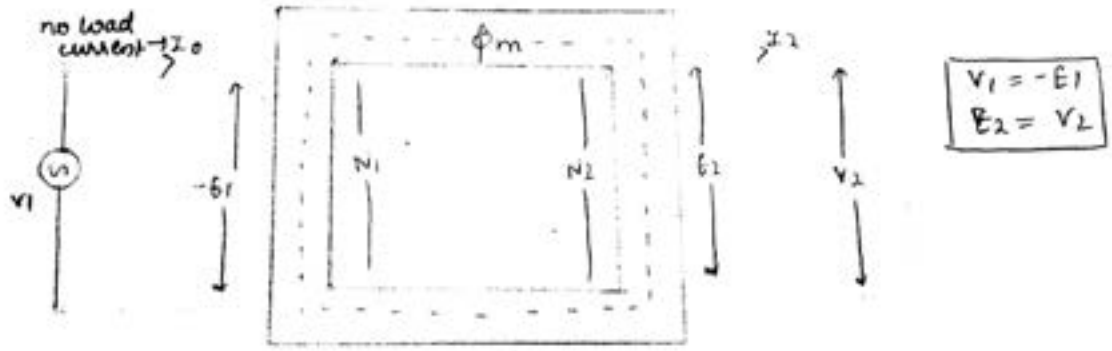
Ideal Transformer :-

An ideal transformer is an imaginary transformer whose efficiency is always 100%

There are four conditions for ideal transformer.

1. Copper loss should be equals to zero it means the primary and secondary winding have zero resistance.
   copper loss = I^2R have to be zero.
2. Iron loss should be equals to zero
3. No leakage between flux in primary and secondary coil that means whole flux is confined to magnetic circuit.
4. No loss due to winding resistance i.e. magnetising current is equals to zero (No loss due to hysteresis & eddy current).

Ideal transformer on No-load :-



No load current (I_0) is equals to 2-5% of full load rated current.

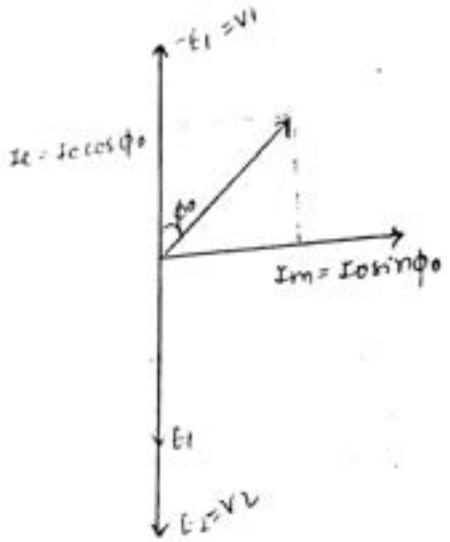
If ideal transformer in no load, i.e, its secondary winding is open circuited.

No load current (I_0)

- I_e (Energy component)
  It is responsible for internal losses
- I_m (Magnetic component)
  It is responsible for flux in core

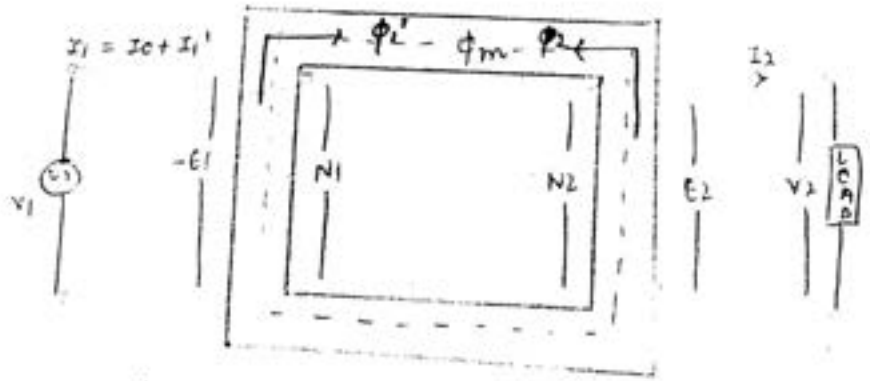
When an alternating voltage V_1 is applied to the primary winding. It draws very small current known as no load current which has two components I_e or I_m. I_e is responsible for internal losses and I_m is responsible to establish the flux in the core. due to AC flux in primary and secondary winding emf (E_2 and $-E_1$) is produced or induced.

$$I_e = I_0 \cos\phi_0$$
$$I_m = I_0 \sin\phi_0$$
$$I_0 = \sqrt{I_m^2 + I_e^2}$$
$$\tan\phi_0 = \frac{I_m}{I_e}$$
$$\boxed{\phi_0 = \tan^{-1}\left[\frac{I_m}{I_e}\right]}$$



Ideal Transformer on load :-

I_1' → counter balancing current



When the load is connected across the terminals of the secondary winding of the ideal transformer, the transformer is said to be loaded and the load current is flows through secondary winding and the load.

$$I_0 N_1 \propto \phi_m$$
$$N_2 I_2 \propto \phi_2$$
$$N_1 I_1' \propto \phi_1'$$
$$\boxed{N_1 I_1' = N_2 I_2}$$

No load current is being neglected because transformer is ideal one

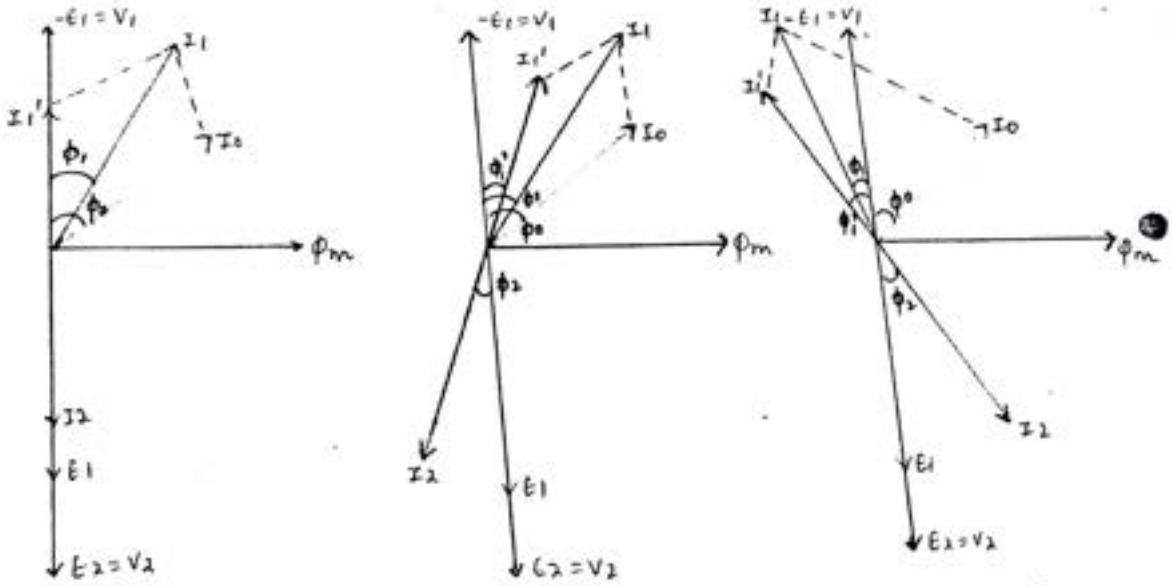
$$\boxed{I_1 \simeq \frac{I_1'}{I_2} = \frac{N_2}{N_1} = \frac{E_2}{E_1} = \frac{V_2}{V_1} = K}$$

- The primary winding must draw enough current to neutralise the demagnetising effect of secondary current so that the main flux in the core remain constants.

Hence mutual flux is constant.

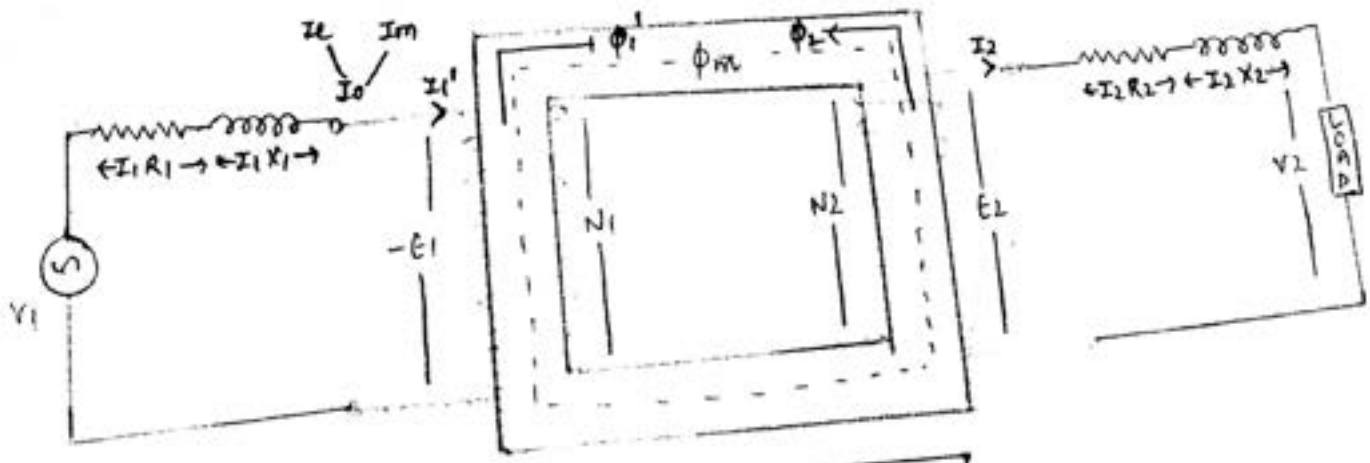
It proof that Transformer is a constant flux machine.

Phasor Diagrams :



1. Resistive (R)
2. Inductive (R-L)
3. capacitive (R-C)

Actual Transformer :



$$V_1 = -E_1 + I_1R_1 + I_1X_1$$

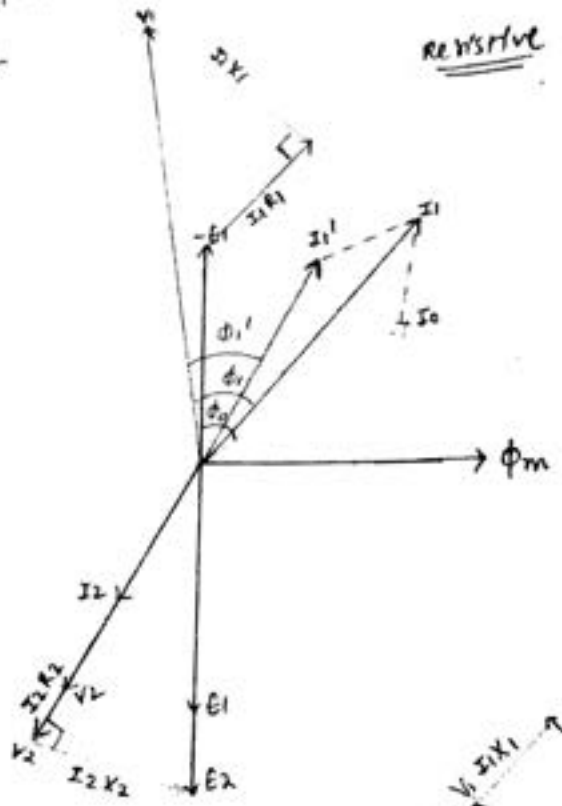
$$E_2 = V_2 + I_2R_2 + I_2X_2$$

The actual transformer has primary and secondary windings of resistance R_1 and R_2 and reactance X_1 and X_2. The impedence of primary winding is $Z_1 = R_1 + jX_1$ and impedence of secondary winding is $Z_2 = R_2 + jX_2$.

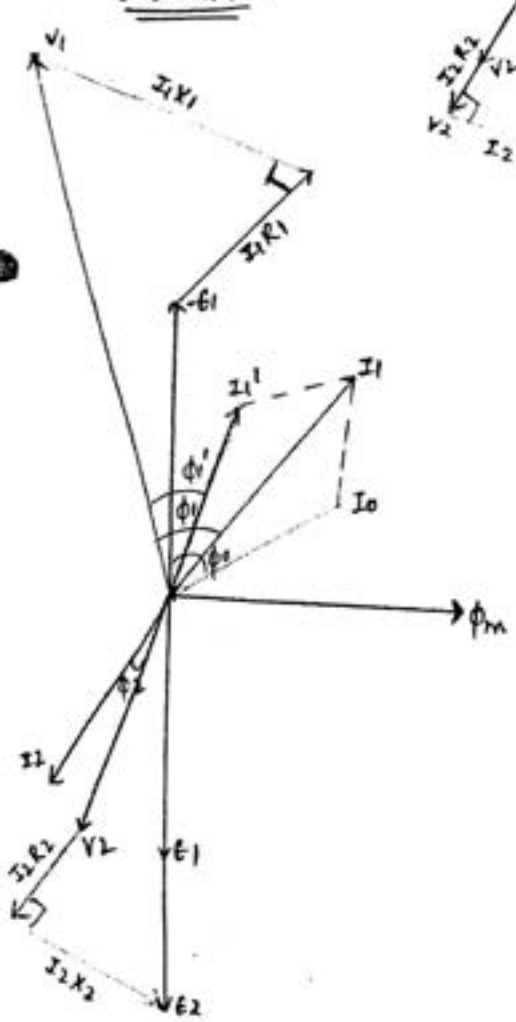
The Actual Transformer has all losses.

Phasor Diagram:

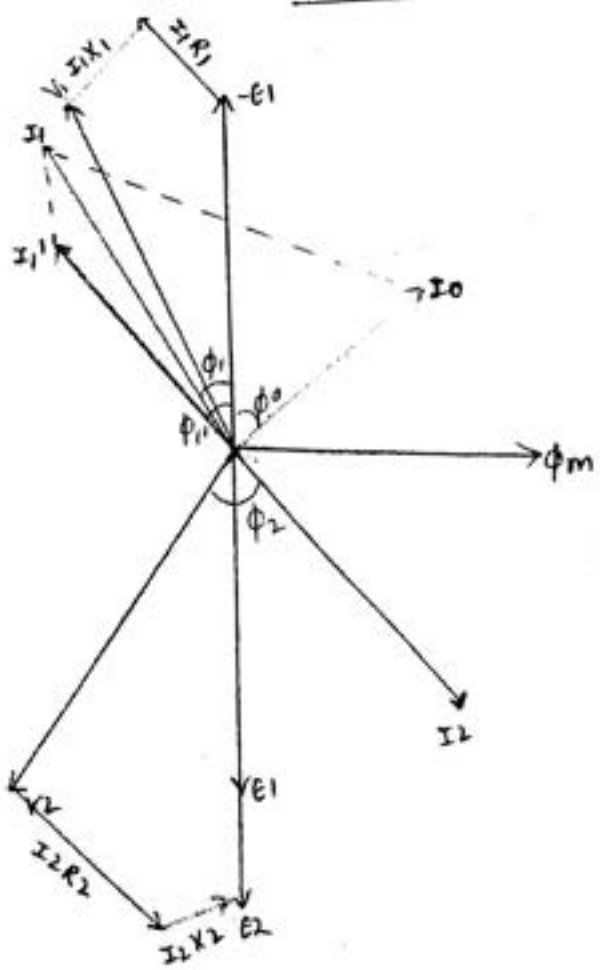
Resistive



Inductive



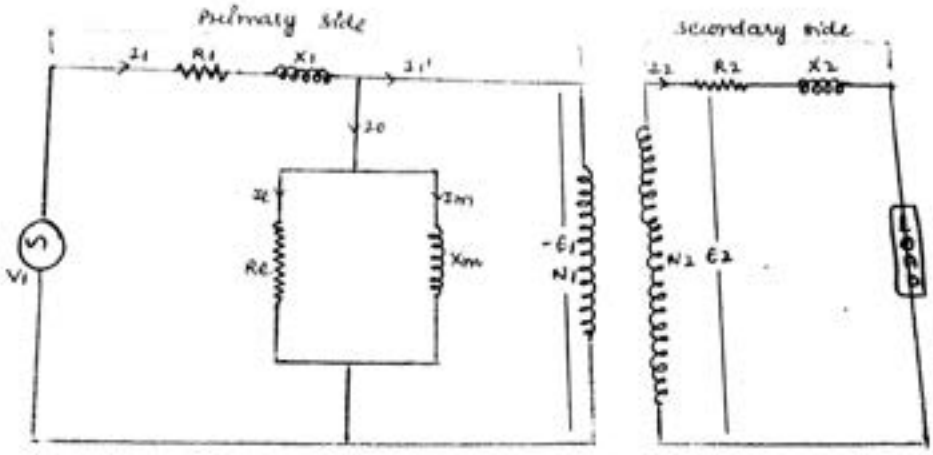
Capacitive



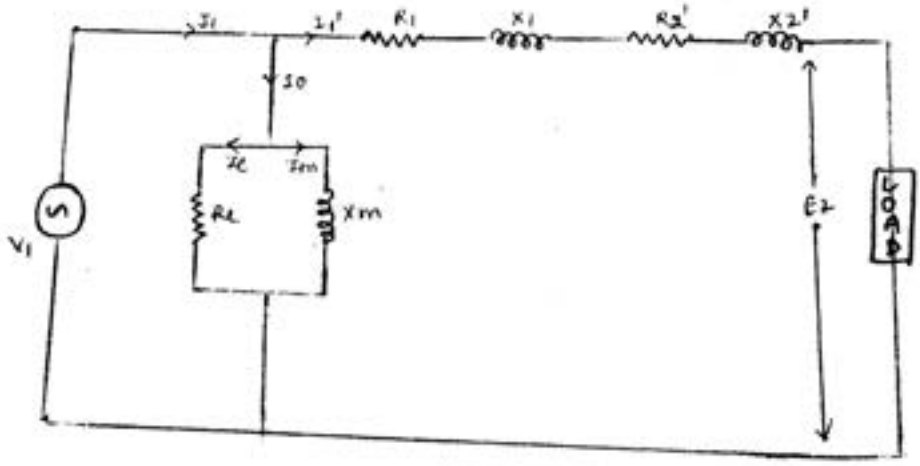
7 NO.

Equivalent circuit of Transformer :-

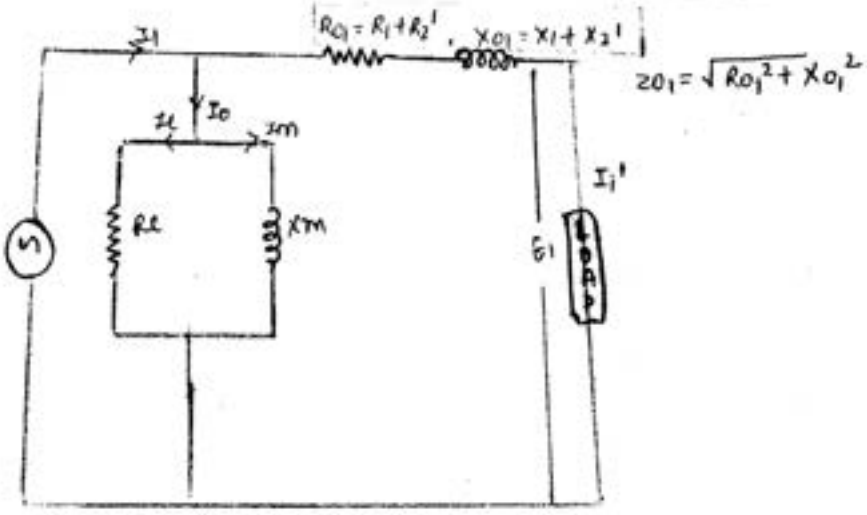
(i)



(ii)



(iii)



Secondary winding refers to primary winding

When the power across primary reactance and resistance (X_1 and R_1) due to current 'I_1' is equal to power across secondary winding reactance and resistance (R'_2 to X'_2) refers to primary side due to 'I_1'.

$$I_2^2 X_2 = I_1^2 X_2'$$

$$X_2' = \left(\frac{I_2}{I_1}\right)^2 X_2$$

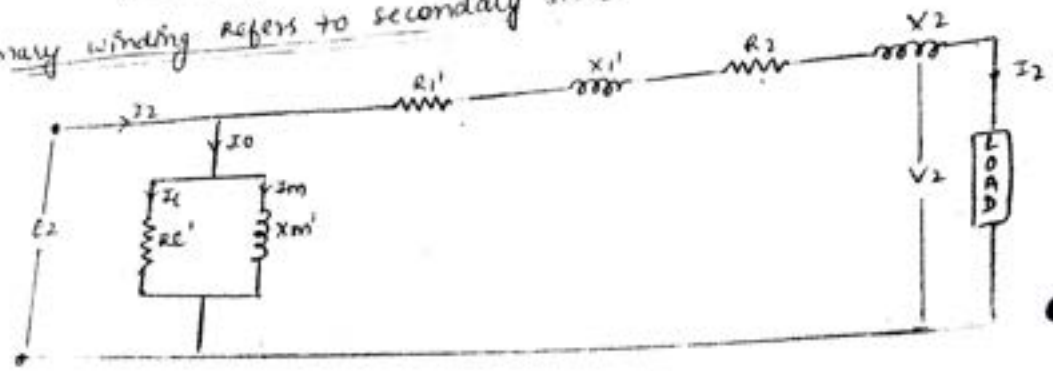
$$\boxed{X_2' = \frac{X_2}{K^2}}$$

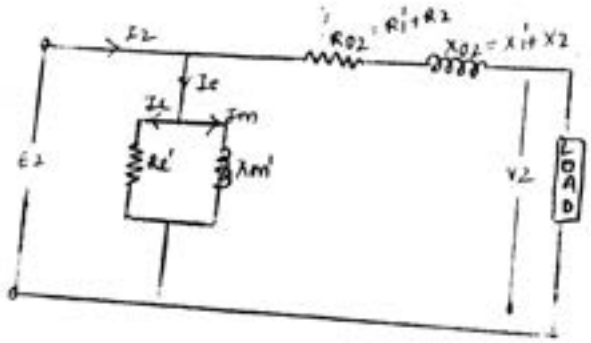
$$I_2^2 R_2 = I_1^2 R_2'$$

$$R_2' = \left(\frac{I_2}{I_1}\right)^2 R_2 \qquad \text{where } \frac{I_1}{I_2} = K$$

$$\boxed{R_2' = \frac{R_2}{K^2}}$$

primary winding refers to secondary side





$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$

$$I_1^2 R_1 = I_2^2 R_1'$$

$$R_1' = \left(\frac{I_1}{I_2}\right)^2 R_1$$

$$\boxed{R_1' = K^2 R_1}$$

$$I_1^2 X_1 = I_2^2 X_1'$$

$$X_1' = \left(\frac{I_1}{I_2}\right)^2 X_1$$

$$\boxed{X_1' = K^2 X_1}$$

when the power across primary reactance and resistance (X_1 and R_1) due to current 'I_1' is equals to the primary resistance and reactance refers to secondary side (R_1' and X_1') due to current I_2.

Ques A 30 KVA, 2000/200 V, single phase, 50 Hz transformer has a primary resistance of 3.5 ohm and reactance 4.5 ohms. The secondary resistance and reactance are 0.015 ohm and 0.02 ohms respectively. Find
i) Equivalent reactance, resistance and impedence referred to the primary side and secondary side ii) Total copper loss.

⇒ Given values

$$V_2 I_2 = V_1 I_1 = 30 \times 10^3 \text{ VA}$$

$$\frac{2000}{200} = \frac{V_1}{V_2}$$

$$K = \frac{V_2}{V_1} = \frac{200}{2000} = 0.1 \quad , f = 50 \text{ Hz}$$

$R_1 = 3.5\,\Omega$ $\quad X_1 = 4.5\,\Omega$ $\quad R_2 = 0.015\,\Omega$ $\quad X_2 = 0.02\,\Omega$

For Primary

$$R_{01} = R_1 + R_2'$$

$$R_2' = \frac{R_2}{K^2} = \frac{0.015}{(0.1)^2} = 1.5\,\Omega$$

$$\boxed{R_2' = 1.5\,\Omega}$$

$$R_{01} = 3.5 + 1.5 = 5\,\Omega$$

$$\boxed{R_{01} = 5\,\Omega}$$

$$X_2' = \frac{X_2}{K^2} = \frac{0.02}{(0.1)^2} = 2\,\Omega$$

$$\boxed{X_2' = 2\,\Omega}$$

$$X_{01} = X_1 + X_2' = 4.5 + 2.6\,\Omega = 6.5\,\Omega$$

$$\boxed{X_{01} = 6.5\,\Omega}$$

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{5^2 + 6.5^2} = 8.2\,\Omega$$

$$\boxed{Z_{01} = 8.2\,\Omega}$$

For secondary

$$R_1 = K^2 R_1 = (0.1)^2 \times 3.5 = 0.035\,\Omega$$

$$\boxed{R_1 = 0.035\,\Omega}$$

$$R_{02} = R_1' + R_2 = 0.035 + 0.015 = 0.05\,\Omega$$

$$\boxed{R_{02} = 0.05\,\Omega}$$

$$X_1' = K^2 X_1 = (0.1)^2 \times 4.5 = 0.045\,\Omega$$

$$\boxed{X_1' = 0.045\,\Omega}$$

$$X_{02} = X_1' + X_2 = 0.045 + 0.02 = 0.065\,\Omega$$

$$\boxed{X_{02} = 0.065\,\Omega}$$

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{0.05^2 + 0.065^2}$$

$$\boxed{Z_{02} = 0.082\,\Omega}$$

Copper loss $\quad P_{cu} = I^2 R$

(i) $I_1^2 R_{01} = 15^2 \times 5 = 1125$ watt

(ii) $I_2^2 R_{02} = 150^2 \times 0.05 = 1125$ watt

Q) The ohmic values of the circuit parameters of a transformer, having a turns ratio of 5, are $R_1 = 0.5\Omega$, $R_2 = 0.021\Omega$, $X_1 = 3.2\Omega$, $X_2 = 0.12\Omega$, $R_c = 350\Omega$ and $X_m = 98\Omega$ referred to primary. Draw the approximate equivalent circuits of the transformer referred to secondary show the numerical values of the circuit parameters.

⇒ Given values

$$\frac{N_1}{N_2} = \frac{5}{1} \quad , \quad K = 0.2$$

$$R_1 = 0.5\Omega \quad , \quad R_2 = 0.021\Omega \quad , \quad X_1 = 3.2\Omega \quad , \quad X_2 = 0.12\Omega$$

$$R_1' = K^2 R_1 = (0.2)^2 \times 0.5 = 0.02$$

$$\boxed{R_1' = 0.02\Omega}$$

$$R_{02} = R_1' + R_2 = 0.02 + 0.021$$

$$\boxed{R_{02} = 0.041\Omega}$$

$$X_1' = K^2 X_1 = (0.2)^2 \times 3.2 = 0.128$$

$$\boxed{X_1' = 0.128\Omega}$$

$$X_{02} = X_1' + X_2 = 0.128 + 0.12$$

$$\boxed{X_{02} = 0.248\Omega}$$

$$R_c' = K^2 R_c = (0.2)^2 \times 350$$

$$\boxed{R_c' = 14\Omega}$$

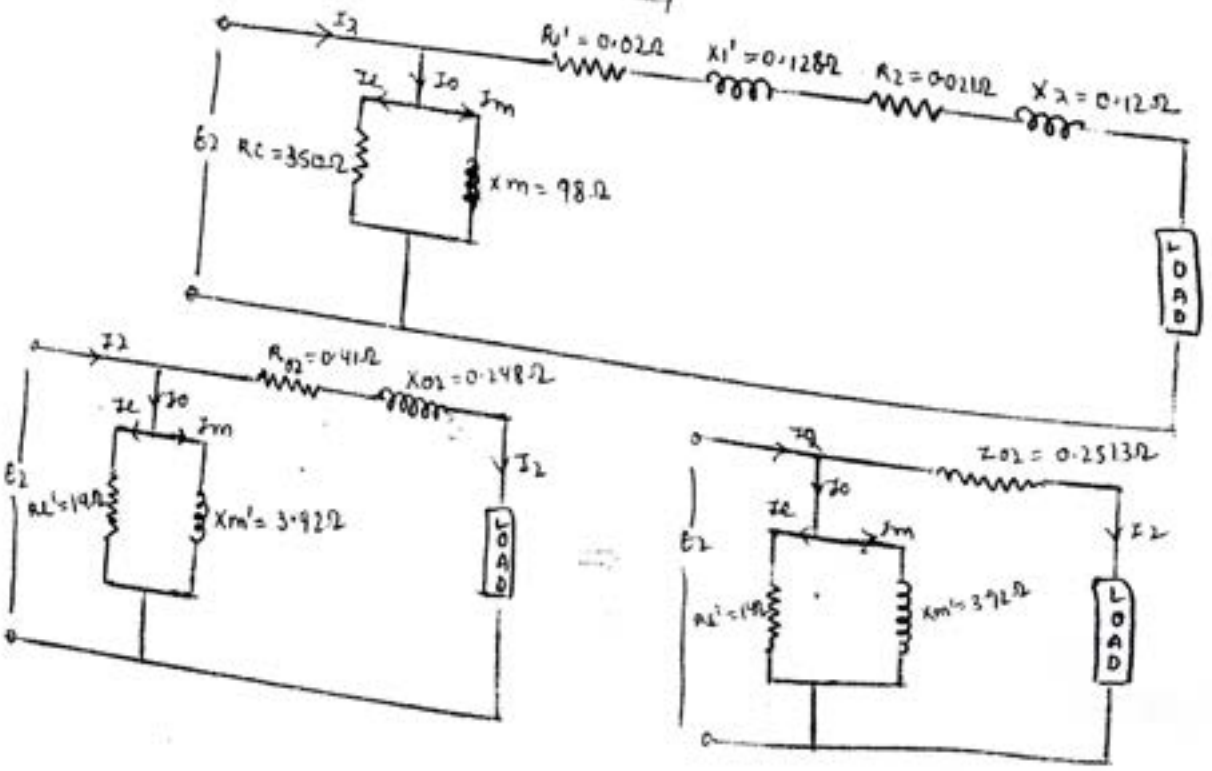
$$X_m' = K^2 X_m = (0.2)^2 \times 98$$

$$\boxed{X_m' = 3.92\Omega}$$

$$Z_{02} = \sqrt{(0.041\Omega)^2 + (0.248)^2}$$

$$= \sqrt{0.063185}$$

$$Z_{02} = 0.2513$$



Transformer losses :

A transformer's output power is always slightly less than transformer's input power. These power losses end up as heat that must be removed from the transformer.

There are two types of transformer losses :

1. Ohmic loss or copper loss (variable loss) :

Copper loss is also known as 'ohmic loss'. Copper loss is already depends on load so it changes with load. Hence it is also known as variable loss.

It is due to alternating current in windings.

It further divided into two :

1. Primary side winding : $P = (I_1^2 R_{01})$
2. Secondary side winding : $P = (I_2^2 R_{02})$

To reduce copper loss we have to minimize the resistance of the wire in the primary and secondary windings and the current flowing through them.

2. Core loss or Iron loss (constant loss) :

Core loss is also known as 'Iron loss'. The value of core loss is always constant. so it is also called constant loss. Do not depend on load current.

It is due to alternating flux in core.

Further divided into two parts :

1. Eddy current loss :

When the AC flux is passing through the magnetic core due to electromagnetic induction law another ~~flux~~ emf is produced in the core. Due to these emf there would be currents which would be in circulating behaviour. These current is called Eddy current. The circulating current will not contribute in output of transformer and dissipated as heat. This type of energy is called Eddy current losses.

$$P_e = \eta_e B^2_{max} f^2 t^2 V \text{ watt}$$

$i_e \uparrow$

we know $\downarrow R \propto \frac{1}{A \uparrow}$

To minimize the eddy current loss we have to decrease the area by laminating core by thin transparent sheet.

2. Hysteresis loss :

During the process of magnetizing and demagnetizing of magnetic substances, energy is spent and this is spent energy appear in the form of heat. The heat loss is known as hysteresis loss.

$$P_h = \eta f B_{max}^2 f^2 V \text{ watt}$$

where $B = \frac{\phi}{A} \Rightarrow$ flux density

t = thickness of transparent sheet

V = volume of the core.

To minimize the hysteresis loss we choose good quality of iron (using material that has less area of the hysteresis loop). BH loop is small.

Imp

Efficiency of Transformer :

$$\% \eta = \frac{\text{output power}}{\text{input power}} \times 100$$

Input Power $= V_1 I_1 \cos\phi_1$

Output Power $= V_2 I_2 \cos\phi_2$

There are two types of losses.

1. Iron loss (P_i)
2. copper loss
   1) Primary side ($I_1^2 R_{01}$)
   2) Secondary side ($I_2^2 R_{02}$)

Input power = output power + losses

Output power = Input power − losses

$$\% \eta = \frac{\text{output power}}{\text{out. power} + \text{losses}}$$

$$\% \eta = \frac{V_2 I_2 \cos\phi_2}{V_2 I_2 \cos\phi_2 + P_i + I_2^2 R_{02}} \times 100$$

$$\% \eta = \frac{\text{i/p power} - \text{losses}}{\text{i/p power}}$$

$$\% \eta = \frac{V_1 I_1 \cos\phi_2 - P_i - I_2^2 R_{01}}{V_1 I_1 \cos\phi_1} \times 100$$

Condition for max. efficiency at full load :

$$\eta = \frac{V_1 I_1 \cos\phi_1 - P_i - I_1^2 R_{01}}{V_1 I_1 \cos\phi_1}$$

$$\frac{d\eta}{dI_1} = 0$$

$$= \frac{d}{I_1}\left[1 - \frac{P_i}{V_1 I_1 \cos\phi_1} - \frac{I_1^2 R_{01}}{V_1 I_1 \cos\phi_1}\right]$$

$$0 = 0 + \frac{P_i}{V_1 I_1^2 \cos\phi_1} - \frac{R_{01}}{V_1 \cos\phi_1}$$

$$0 = \frac{P_i}{I_1^2} - R_{01}$$

$$P_i = I_1^2 R_{01}$$

$$P_i = P_{cu}$$

Q06 paper

Condition for Efficiency at variable load :

Let x is a function by which load is varying.

for eg : at half load $x = 1/2 = 0.5$

at 75% of full load $x = 3/4 = 0.75$

at $1/4^{th}$ load $= \frac{1}{4} = 0.25$

at full load $= 1$

$$\% \eta = \frac{x V_2 I_2 \cos\phi_2}{x V_2 I_2 \cos\phi_2 + P_i + x^2 P_{cu[FL]}}$$

$$\eta = \frac{x V_1 I_1 \cos\phi_1 - P_i - x^2 I_1^2 R_{01}}{x V_1 I_1 \cos\phi_1}$$

$$\frac{d\eta}{dx} = 0 = \frac{d}{dx}\left[1 - \frac{P_i}{x V_1 I_1 \cos\phi_1} - \frac{x^2 I_1^2 R_{01}}{x V_1 I_1 \cos\phi_1}\right]$$

$$0 = 0 + \frac{P_i}{x^2 V_1 I_1 \cos\phi_1} - \frac{I_1 R_{01}}{V_1 \cos\phi}$$

$$0 = \frac{P_i}{x^2 I_1} - I_1 R_{01}$$

$$P_i = x^2 I_1^2 R_{01}$$

$$\boxed{P_i = x^2 P_{cu}}$$

$$\boxed{x = \sqrt{\frac{P_i}{P_{cu}}}}$$

(at full load $x = 1$)

$1.25 \rightarrow 0.25$
$80 \rightarrow 0.90$

* where $x = N$

copper losses for variable load $= x^2 P_{cu}$ [F.L]

$$\boxed{\% \eta_{max} = \frac{VA \text{ for } \eta_{max} \times 10^3 \times \cos\phi \times 100}{VA \text{ for } \eta_{max} \times 10^3 \times \cos\phi + 2P_i}}$$

$$\boxed{\text{Corresponding load for KVA for max Efficiency} = \text{Rated KVA} \times \sqrt{\frac{P_i}{P_{cu}}}}$$

Voltage Regulation:

Voltage regulation may be defined as it is the change in voltage across the secondary terminal of the transformer from no load to full load.

$$\text{Voltage Regulation} = \frac{\text{No load voltage} - \text{full load voltage}}{\text{No load Rated voltage}}$$

$$\boxed{\% VR = \frac{E_2 - V_2}{E_{2[Rated]}} \times 100}$$

Zero voltage regulation will be found ~~ound~~ out at capacitive load

$$E_2 = V \pm (I_2 R_2 \cos\phi + I_2 X_2 \sin\phi)$$

$$\boxed{VR = \frac{I_2 [R_{02}\cos\phi_2 \pm X_{02}\sin\phi_2]}{V_2}}$$

→ (+) lagging P.F

(−) leading P.F

Voltage regulation will be at leading power factor or capacitive load.

Ques 3 A 250 KVA single phase transformer has iron loss of 1.8 KW. The full load copper loss is 2000 watts. Calculate (i) Efficiency at full load 0.8 lagging p.f (ii) KVA supplied at max. efficiency (iii) maximum efficiency at 0.8 lagging p.f.

⇒ Given values

$V_2 I_2 = V_1 I_1 = 250 \times 10^3$ VA

$P_i = 1.8$ KW $= 1800$ watt

P_{cu} [F.L] $= 2000$ watt

(i) $\eta_{(F.L)}$, $\cos\phi = 0.8$ [lag], $x = 1$

$$\eta_{F.L} = \frac{x V_2 I_2 \cos\phi_2}{x V_2 I_2 \cos\phi_2 + P_i + x^2 P_{cu(F.L)}} \times 100$$

$$= \frac{1 \times 250 \times 10^3 \times 0.8}{1 \times 250 \times 10^3 \times 0.8 + 1800 + (1)^2 \times 2000} \times 100$$

$$= \frac{2 \times 10^5}{2 \times 10^5 + 1800 + 2000} \times 100$$

$$= \frac{200000}{203800} \times 100$$

$$\boxed{\% \eta = 98.135\%}$$

(ii) KVA at η_{max} or corresponding load for η_{max} = Rated KVA $\sqrt{\frac{P_i}{P_{cu}}}$

$$= 250 \sqrt{\frac{1800}{2000}}$$

$$= 250 \times 0.94$$

$$\boxed{= 237.17 \text{ KVA}}$$

(iii) Maximum Efficiency at $\cos\phi = 0.8$

$$\% \eta_{max} = \frac{\text{VA for } \eta_{max} \times 10^3 \times \cos\phi}{\text{VA for } \eta_{max} \times 10^3 \times \cos\phi + 2P_i} \times 100$$

$$= \frac{237.17 \times 10^3 \times 0.8}{237.17 \times 10^3 \times 0.8 + 2 \times 1800} \times 100$$

$$\boxed{\% \eta_{max} = 98.137\%}$$

Ques 4 The efficiency of a 400KVA, single phase transformer is 98.77% at full load 0.8 pf and 99.13% at half load unity p.f. Find (A) Iron losses at full and half full loads (B) Copper losses at full and half off full loads.

⇒ Given values

$\eta_1 = 99.13\%$, $\cos\phi_1 = 1$, $x_1 = 0.5$

$\eta_2 = 98.13\%$, $\cos\phi_2 = 0.8$, $x_2 = 1$ (Data)

$$\% \eta = \frac{x_1 V_2 I_2 \cos\phi_1}{x_1 V_2 I_2 \cos\phi_2 + P_i + x_1^2 P_{cu}} \times 100$$

$$99.13 = \frac{0.5 \times 400 \times 10^3 \times 100}{0.5 \times 400 \times 10^3 + P_i + 0.25 P_{cu}}$$

$$2 \times 10^4 + P_i + 0.25 P_{cu} = \frac{2 \times 10^4}{99.13}$$

$$P_i + 0.25 P_{cu} = 1755.27 \quad \text{—— (1)}$$

$$\% \eta = \frac{x_2 V_2 I_2 \cos\phi_2 \times 100}{x_2 V_2 I_2 \cos\phi_2 + P_i + x_2^2 P_{cu}}$$

$$98.77 = \frac{400 \times 10^3 \times 0.8 \times 100}{400 \times 10^3 \times 0.8 + P_i + P_{cu}}$$

$$32 \times 10^4 + P_i + P_{cu} = \frac{32 \times 10^4}{98.77}$$

$$P_i + P_{cu} = 3985.01 \quad \text{—— (2)}$$

from (1) and (2)

$$P_i + 0.25 P_{cu} = 1755.27$$
$$P_i + P_{cu} = 3985.01$$

$$\boxed{P_i = 1012.0233 \text{ watt}}$$
$$\boxed{P_{cu} = 2972.9866 \text{ watt}}$$

P_{cu} on half load $= x^2 P_{cu(FL)}$

$$= (0.5)^2 \times 2972.9866$$

$$P_{cu(HL)} = 743.24665 \text{ watt}$$

Ques 5 The maximum efficiency of a 100KVA transformer is 98.48% and operate at 90% full load unity power factor. Calculate the efficiency of a transformer at unity power factor at full load.

Given values

$$V_2 I_2 = V_1 I_1 = 100 \times 10^3 \text{ VA}$$

$$\eta_{max} = 98.48\%$$

$$x = 0.9$$

$$\cos\phi = 1$$

$\eta = ?$, $x = 1$, $\cos\phi = 1$

$$\% \eta_{max} = \frac{\text{VA for } \eta_{max} \times 10^3 \times \cos\phi \times}{\text{VA for } \eta_{max} \times 10^3 \times \cos\phi +}$$

$$0.9848 = \frac{0.9 \times 100 \times 10^3}{0.9 \times 100 \times 10^3 + 2P_i}$$

$$0.9848 = \frac{9 \times 10^4}{9 \times 10^4 + 2P_i}$$

$$2P_i = 91389.114 - 90000$$

$$2P_i = 1389.114$$

$$P_i = 694.557$$

$$0.9848 = \frac{0.9 \times 100 \times 10^3}{0.9 \times 100 \times 10^3 + 694.55 + (0.9)^2 P_{cu}}$$

$$(0.9)^2 P_{cu} = 91389.114 - 90694.55$$

$$P_{cu} = \frac{694.564}{0.81}$$

$$\boxed{P_{cu} = 857.486}$$

$$P_{cu(HL)} = x^2 P_{cu(FL)}$$

$$\% \eta_{FL} = \frac{1 \times 100 \times 10^3 \times 1 \times 100}{1 \times 100 \times 10^3 \times 1 + 694.55 + 857.486}$$

$$\boxed{\% \eta_{FL} = 98.3912\%}$$

Ques The maximum efficiency of a 100KVA, 1100/440 volt 50Hz transformer is 96%. This occurs at 75% of full load at 0.8 pf lagging. Find the efficiency of Transformer at 3/4 F.L at 0.6 pf leading.

⇒ given values

$$V_2 I_2 = V_1 I_1 = 100 \times 10^3 \text{ VA}$$

$$\frac{V_1}{V_2} = \frac{1100}{400} \text{ V}$$

$$f = 50 \text{ Hz}$$

$$\eta_{max}\% = \frac{\text{KVA at } \eta_{max} \times \cos\phi \times 10^3}{10^3 \times \text{KVA at } \eta_{max} \times \cos\phi + 2P_i}$$

$$0.96 = \frac{100 \times 10^3 \times 0.8}{100 \times 10^3 \times 0.8 + 2P_i}$$

$$0.96 = \frac{80000}{80000 + 2P_i}$$

$$2P_i = 83333.33 - 80000$$

$$P_i = \frac{3333.33}{2} = 1666.665$$

For η_{max} $P_i = P_{cu}$

$$\eta\% = \frac{x V_2 I_2 \cos\phi_2}{x V_2 I_2 \cos\phi_2 + P_i + x^2 P_{cu}} \times 100$$

$$\eta\% = \frac{0.75 \times 100 \times 10^3 \times 0.6 \times 100}{0.75 \times 100 \times 10^3 \times 0.6 + 1666.6 + (0.75)^2 1666.6}$$

$$= \frac{45 \times 10^5}{45 \times 10^3 + 1666.6 + 937.4625}$$

$$= \frac{45 \times 10^5}{47604.062}$$

$$\boxed{\eta\% = 94.5\%}$$

Unit-3 Single Phase Transformer

Important formulae used in numericals

(i) Emf Equation $\boxed{E_{rms} = 4.44 N \phi . f}$

$N \rightarrow$ No. of turns

(ii) secondary winding refers to primary winding

$\boxed{R_2' = \frac{R_2}{K^2}}$ $\boxed{X_2' = \frac{X_2}{K^2}}$ $\boxed{R_{01} = R_1 + R_2'}$ $\boxed{X_{01} = X_1 + X_2'}$

$\boxed{Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}}$

(iii) Primary winding refers to secondary winding

$\boxed{R_1' = K^2 R_1}$ $\boxed{X_1' = K^2 X_1}$ $\boxed{R_{02} = R_1' + R_2}$ $\boxed{X_{02} = X_1' + X_2}$

$\boxed{Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}}$

(iv) Efficiency of Transformer

$$\boxed{\%\eta = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + P_i + I_2^2 R_{02}} \times 100}$$

$$\boxed{\%\eta = \frac{V_1 I_1 \cos\phi - P_i - I_2^2 R_{01}}{V_1 I_1 \cos\phi_1} \times 100}$$

(v) condition for maximum efficiency $\boxed{P_i = P_{cu}}$

(v) condition at variable load $\boxed{P_i' = x^2 P_{cu}}$

$$\boxed{x = \sqrt{\frac{P_i'}{P_{cu}}}}$$

(vi) For maximum efficiency

$$\boxed{\%\eta_{max} = \frac{VA \text{ for } \eta_{max} \times 10^3 \times \cos\phi \times 100}{VA \text{ for } \eta_{max} \times 10^3 \times \cos\phi + 2P_i}}$$

(vii) $\boxed{\text{Corresponding load OR KVA for max. Efficiency} = \text{Rated KVA} \sqrt{\frac{P_i}{P_{cu}}}}$

For star system

(I) $V_L = \sqrt{3} V_{ph}$

(II) $I_L = I_{ph}$

(III) $P = \sqrt{3} V_L I_L \cos\phi$ watt

(IV) $Q = \sqrt{3} V_L I_L \sin\phi$ VAR

(V) $S = \sqrt{3} V_L I_L$ VA

(VI) $\cos\phi = \frac{R_{ph}}{Z_{ph}}$

(VII) $Z_{ph} = \frac{V_{ph}}{I_{ph}}$

(VIII) $Z_{ph} = \sqrt{R_{ph}^2 + X_{ph}^2}$

For Delta system

(I) $V_L = V_{ph}$

(II) $I_L = \sqrt{3} I_{ph}$

(III) $P = \sqrt{3} V_L I_L \cos\phi$ Watt

(IV) $Q = \sqrt{3} V_L I_L \sin\phi$ VAR

(V) $S = \sqrt{3} V_L I_L$ VA

(VI) $\cos\phi = \frac{R_{ph}}{Z_{ph}}$

(VII) $Z_{ph} = \frac{V_{ph}}{I_{ph}}$

(VIII) $Z_{ph} = \sqrt{R_{ph}^2 + X_{ph}^2}$

Ques A 1100/110V, 22KVA, single phase transformer has primary resistance and reactance 4Ω and 6Ω respectively. The secondary resistance and reactance are 0.04Ω and 0.065Ω

(i) Equivalent resistance and reactance of secondary referred to primary.
(ii) Total resistance and reactance referred to primary.
(iii) Equivalent resistance and reactance of primary referred to secondary.
(iv) Total copper loss.

Ans Given values $\frac{V_1}{V_2} = \frac{1100}{110}$, $V_1I_1 = V_2I_2 = 22\times10^3 VA$, $K = \frac{V_2}{V_1} = 0.1$

$R_1 = 4\Omega$, $X_1 = 6\Omega$, $R_2 = 0.04\Omega$, $X_2 = 0.065\Omega$

(i) For secondary referred to primary

$R_2' = \frac{R_2}{K^2} = \frac{0.04}{(0.1)^2} = 4\Omega$ $\boxed{R_2' = 4\Omega}$

$X_2' = \frac{X_2}{K^2} = \frac{0.065}{(0.1)^2} = 6.5\Omega$ $\boxed{X_2' = 6.5\Omega}$

(ii) Total resistance and reactance referred to primary

$R_{01} = R_1 + R_2' = 4 + 4\Omega = 8\Omega$ $\boxed{R_{01} = 8\Omega}$

$X_{01} = X_1 + X_2' = 6 + 6.5\Omega = 12.5\Omega$ $\boxed{X_{01} = 12.5\Omega}$

(iii) For primary referred to secondary

$R_1' = K^2R_1 = (0.1)^2 \times 4 = 0.04\Omega$ $\boxed{R_1' = 0.04\Omega}$

$X_2' = K^2X_2 = (0.1)^2 \times 6 = 0.06\Omega$ $\boxed{X_1' = 0.06\Omega}$

$R_{02} = R_2 + R_1' = 0.04 + 0.04 = 0.08\Omega$

(iv) Total copper loss

$22\times10^3 = V_1I_1$

$22\times10^3 = 1100 \cdot I_1$

$\boxed{20A = I_1}$

$22000 = V_2I_2$

$22000 = 110 \cdot I_2$

$\boxed{200A = I_2}$

$P_{cu1} = I_1^2R_{01} = (20)^2 \times 8 = 3200W$

$\boxed{P_{cu1} = 3200W}$

$P_{cu2} = I_2^2R_{02} = (200)^2 \times 0.08 = 3200W$

$\boxed{P_{cu2} = 3200W}$

Q1 A transformer is rated at 100 KVA. At full load its copper loss is 1200W and iron losses are 960W. Calculate (i) efficiency at half load, 0.8 pf lagging (ii) efficiency at full load, unity p.f (iii) efficiency at 75% full load, 0.7 pf lagging (iv) the load KVA at which maximum efficiency occurs (v) The maximum at 0.85 pf lagging.

Ans Given values $V_1I_1 = V_2I_2 = 100 \times 10^3 VA$

(i) $\eta = ?$, $x = 0.5$, $\cos\phi = 0.8$

$$\eta\% = \frac{0.5 \times 100 \times 10^3 \times 0.8 \times 100}{0.5 \times 100 \times 10^3 \times 0.8 + 960 + (0.5)^2 \times 1200}$$

$$= \frac{4000000}{41260} = 96.94\%$$

$$\boxed{\eta\% = 96.94\%}$$

(ii) $\eta = ?$, $x = 1$, $\cos\phi = 1$

$$\eta\% = \frac{x V_2 I_2 \cos\phi}{x V_2 I_2 \cos\phi + P_i + x^2 Cu} \times 100$$

$$= \frac{100 \times 10^3 \times 100}{100 \times 10^3 + 960 + 1200} = \frac{10^7}{102160} = 97.88\%$$

$$\boxed{\eta\% = 97.88\%}$$

(iii) $\eta = ?$, $x = 0.75$, $\cos\phi = 0.7$

$$\eta\% = \frac{0.75 \times 100 \times 10^3 \times 0.7 \times 100}{0.75 \times 100 \times 10^3 \times 0.7 + 960 + (0.75)^2 \times 1200}$$

$$= \frac{5250000}{54135} = 96.97$$

$$\boxed{\eta\% = 96.97\%}$$

(iv) load KVA at max. η occurs = Rated KVA $\sqrt{\frac{P_i}{P_{cu}}}$

$$= 100\sqrt{\frac{960}{1200}} = 100\sqrt{0.8}$$

$$\boxed{= 89.44 KVA}$$

v) max. efficiency at 0.85 pf

$$\%\eta_{max} = \frac{100 \times 10^3 \times 0.85 \times 100}{100 \times 10^3 \times 0.85 + 2 \times 960}$$

$$= \frac{8500000}{86920}$$

$$\boxed{\%\eta_{max} = 97.79\%}$$

(ii) $\eta_{max} = \frac{25 \times 10^3 \times 0.8}{25 \times 10^3 \times 0.8 + 2 \times 1500} = \frac{2000000}{23000} = 86.95\%$

$$\boxed{\eta_{max} = 86.95\%}$$

(iii) Rated KVA for max efficiency = Rated KVA $\sqrt{\frac{P_i}{P_{cu}}}$

$$= 25\sqrt{\frac{1500}{1800}} = 25\sqrt{0.8}$$

$$\boxed{= 22.54 \text{ KVA}}$$

For max. efficiency $x = 1$, $\cos\phi = 1$

$$\eta\% = \frac{22.54 \times 10^3 \times 100}{22.54 \times 10^3 \times 100 + 3000}$$

$$\boxed{\eta\% = 88.25\%}$$

Ques A single phase, 50Hz, core type transformer has square cores of 20cm side, permissible maximum flux density is 1Wb/m². Calculate the number of turns per limb on the high and low voltage side for 300/220V ratio.

Soln

$f = 50Hz$ $B = 1 Wb/m^2$

$E_1 = 300V$ $E_2 = 220V$

$$E_1 = 4.44 N_1 \phi_m f$$

$$N_1 = \frac{E_1}{4.44 \times B \times A \times f} = \frac{300}{4.44 \times 1 \times 0.04 \times 50}$$

$$\boxed{N_1 = 33.78}$$

$$N_2 = \frac{E_2}{4.44 \times B \times A \times f} = \frac{220}{4.44 \times 1 \times 0.04 \times 50}$$

$$\boxed{N_2 = 24.77}$$

High voltage turns = 34 turns

Low voltage turns = 25 turns

Ques In a 25KVA, 2000V/200V transformer the iron and copper losses are 200W and 400W respectively. Calculate the efficiency at half load and 0.8 pf lagging. Also determine the maximum efficiency and corresponding load.

Ques 3 The maximum efficiency of a 100KVA, 1100/440V, 50Hz transformer is 96%, this occurs at 75% of full load at 0.8 pf lagging. Find the efficiency of transformer at 3/4 full load at 0.6 pf leading.

Soln Given values $V_1I_1 = V_2I_2 = 100\times10^3 VA$, $f = 50Hz$, $\frac{V_1}{V_2} = \frac{1100}{440}V$

$\eta = 96\%$, $x = 0.75$, $\cos\phi = 0.8$

$$\eta_{max} = \frac{VA \text{ for } \eta_{max} \times 10^3 \times \cos\phi \times 100}{VA \text{ for } \eta_{max} \times 10^3 \times \cos\phi + 2P_i}$$

$$0.96 = \frac{100\times10^3\times0.8}{100\times10^3\times0.8 + 2P_i}$$

$$2P_i = 8.33\times10^4 - 8\times10^4$$

$$P_i = \frac{3333}{2} = 1666.5$$

$$\boxed{P_i = 1666.5\,W}$$

for maximum condition of efficiency $\boxed{P_i = P_{cu}}$

$$\eta\% = \frac{0.75\times100\times10^3\times0.6\times100}{0.75\times100\times10^3\times0.6 + 1666.6 + (0.75)^2\times1666.5}$$

$$= \frac{45\times10^5}{47607.062} = 94.5\%$$

$$\boxed{\eta\% = 94.5\%}$$

Ques A 25KVA, 2000/200V transformer has full load copper & iron loss are 1.8KW & 1.5KW respectively. Find (i) The efficiency at half the rated KVA & at unity power factor (ii) The efficiency at full load & at 0.8 pf lagging (iii) KVA load for maximum efficiency & value of max. efficiency

Soln Given values $V_1I_1 = V_2I_2 = 25\times10^3 VA$, $\frac{V_1}{V_2} = \frac{2000}{200}V$, $x = 1$,

$P_{cu} = 1800W$, $P_i = 1500W$

(i) $$\eta\% = \frac{0.5\times25\times10^3\times1\times100}{0.5\times25\times10^3\times1 + 1500 + (0.5)^2\times1800}$$

$$= \frac{1250000}{14450} = 86.50\%$$

$$\boxed{\eta\% = 86.5\%}$$

Ques A 30KVA, 2000/200V, single phase, 50Hz transformer has a primary resistance of 3.5Ω and reactance of 4.5Ω. The secondary resistance and reactance are 0.015Ω and 0.02Ω respectively. Find (i) equivalent resistance, reactance and impedence referred to the primary side (ii) Total copper loss in the transformer.

Soln Given values

$V_1I_1 = V_2I_2 = 30\times10^3 VA$, $\frac{V_1}{V_2} = \frac{2000}{200}$

$\frac{V_2}{V_1} = K = 0.1$, $f = 50Hz$

$R_1 = 3.5\Omega$, $X_1 = 4.5\Omega$, $R_2 = 0.015\Omega$ $X_2 = 0.02\Omega$.

Secondary referred to primary side.

$R_2' = \frac{R_2}{K^2} = \frac{0.015}{(0.1)^2} = 1.5\Omega$ $\boxed{R_2' = 1.5\Omega}$

$X_2' = \frac{X_2}{K^2} = \frac{0.02}{(0.1)^2} = 2\Omega$ $\boxed{X_2' = 2\Omega}$

$R_{01} = R_1 + R_2' = 3.5 + 1.5 = 5\Omega$ $\boxed{R_{01} = 5\Omega}$

$X_{01} = X_1 + X_2' = 4.5 + 2.6 = 6.5\Omega$ $\boxed{X_{01} = 6.5\Omega}$

$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{(5)^2 + (6.5)^2} = 8.2\Omega$

$\boxed{Z_{01} = 8.2\Omega}$

Total copper loss

$P_{cu} = I_1^2 R_{01}$

$= 15^2 \times 5 = 1125W$

$\boxed{P_{cu} = 1125W}$

Ques 55 The emf per turn of a single phase 10KVA, 2200/220V, 50Hz transformer is 10V calculate (i) the no. of turns (ii) the net cross-sectional area of core for max flux density of 1.5T

Soln Given values

Emf per turn = 10V

primary emf $E_1 = V_1 = 2200V$

secondary emf $E_2 = V_2 = 220V$

(i) $N_1 = ?$ $N_1 = \frac{E_1}{\text{Emf per turn}} = \frac{2200}{10} = 220$ turns

$\boxed{N_1 = 220 \text{ turns}}$

Ans: Given values $V_1I_1 = V_2I_2 = 25 \times 10^3 VA$, $\frac{V_1}{V_2} = \frac{2000}{200} V$

$P_{cu} = 400W$, $P_i = 200W$

$x = 0.5$, $\cos\phi = 0.8$

(i) $$\eta = \frac{0.5 \times 25 \times 10^3 \times 0.8 \times 100}{0.5 \times 25 \times 10^3 \times 0.8 + 200 + (0.5)^2 \times 400}$$

$$= \frac{10000 \times 100}{10000 + 200 + 100} = \frac{1000000}{10300}$$

$$\boxed{\eta\% = 97.08\%}$$

(ii) $$\%\eta_{max} = \frac{VA \text{ for } \eta_{max} \times \cos\phi \times 100}{VA \text{ for } \eta_{max} \times \cos\phi + 2P_i}$$

$$= \frac{25 \times 10^3 \times 1 \times 100}{25 \times 10^3 + 2 \times 200} = \frac{25000000}{25400}$$

$$\boxed{\%\eta_{max} = 98.42\%}$$

(iii) Corresponding load = Rated KVA $\sqrt{\frac{P_i}{P_{cu}}}$

$$= 25\sqrt{\frac{200}{400}} = 25 \times 0.7071$$

$$\boxed{= 17.67 KVA}$$

Que: It is desired to have a 4.13 mWb maximum core flux in a transformer at 110V and 50Hz. Determine the required number of turns in primary.

Ans: Given values

$\phi = 4.13 mWb = 4.13 \times 10^{-3} Wb$

$V = 110V$, $f = 50Hz$

$$\boxed{E_{rms} = 4.44 N \phi_0 f}$$

$$N = \frac{E}{4.44 \times \phi_0 \times f} = \frac{110 \times 10^3}{4.44 \times 4.13 \times 50}$$

$$= 119.97$$

$$\boxed{N = 120 \text{ turns}}$$

$$N_2 = \frac{E_2}{\text{Emf per turns}} = \frac{220}{10} = 22 \text{ turns}$$

$$\boxed{N_2 = 22 \text{ turns}}$$

max flux

$$\phi = \frac{Emf}{4.44 \times f} = \frac{10}{4.44 \times 50}$$

$$\boxed{\phi = 0.045 \text{ Wb}}$$

Net cross Area

$$a = \frac{\phi}{B} = \frac{0.045}{1.5} = 0.03 \text{ m}^2$$

$$\boxed{a = 0.03 \text{ m}^2}$$

Ques A transformer rated 200/50V, 10KVA has a core loss of 100 watts. What is the maximum efficiency of the transformer at 0.8 lagging power factor? Assume full load copper loss as 200 watt. & what load this maximum efficiency is obtained?

Soln: Given values $\frac{V_1}{V_2} = \frac{200}{50}$ V, $V_1 I_1 = V_2 I_2 = 10 \times 10^3$ VA, $P_i = 100$ W

$\eta_{max} = ?$, $\cos\phi = 0.8$

$$\% \eta_{max} = \frac{\text{VA for } \eta_{max} \times 10^3 \times 100 \times \cos\phi}{\text{VA for } \eta_{max} \times 10^3 \times \cos\phi + 2P_i}$$

$$= \frac{10 \times 10^3 \times 0.8 \times 100}{10 \times 10^3 \times 0.8 + 200} = \frac{800000}{8200}$$

$$\boxed{\% \eta_{max} = 97.56\%}$$

Assume $P_{cu} = 200$ W.

For variable load

$$P_i = x^2 P_{cu}$$

$$x = \sqrt{\frac{P_i}{P_{cu}}} = \sqrt{\frac{100}{200}}$$

$$= \sqrt{0.5} = 0.707$$

$$\boxed{x = 0.707}$$

Ques In a 25kVA, 2000/200V transformer, the iron and full load copper losses are 350W and 400W respectively. Calculate the efficiency at full load and 0.8 power factor lagging. Determine the max. efficiency and corresponding load.

Soln Given values $V_1I_1 = V_2I_2 = 25 \times 10^3$ VA, $\frac{V_1}{V_2} = \frac{2000}{200}$ V

$P_i = 350$ W $\quad P_{cu} = 400$ W

$$\%\eta = \frac{25 \times 10^3 \times 0.8 \times 100}{25 \times 10^3 \times 0.8 + 350 + 400}$$

$$= \frac{2000000}{20750} = 96.38\%$$

$$\boxed{\%\eta = 96.38\%}$$

Corresponding load = Rated KVA $\sqrt{\frac{P_i}{P_{cu}}}$

$$= 25\sqrt{\frac{350}{400}} = 25 \times 0.9354$$

$$\boxed{= 23.38 \text{ kVA}}$$

$$\%\eta_{max} = \frac{25 \times 10^3 \times 0.8 \times 100}{25 \times 10^3 \times 0.8 + 2 \times 350}$$

$$= \frac{2000000}{20700}$$

$$\boxed{\%\eta_{max} = 96.61\%}$$

Ques A balanced star connected load of $(8+j6)\Omega$ per phase is connected to a 3-phase - 400V supply. Find the line current, power factor, 3-phase power and 3-phase volt-amperes. Also draw the phasor diagram.

Soln Given values $Z = (8+j6)\Omega = \sqrt{8^2+6^2} = 10\Omega$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 231\text{V}$$

① $I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{231}{10} = 23.1$ A

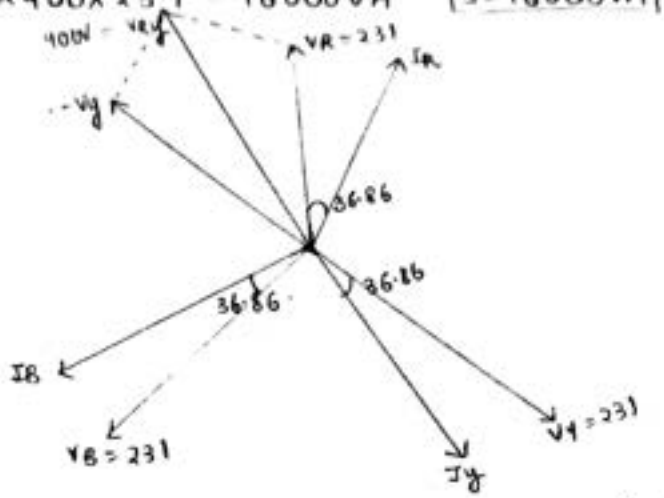
$I_{ph} = I_L \quad \boxed{I_L = 23.1\text{A}}$

(ii) $\cos\phi = \frac{R_{ph}}{Z_{ph}} = \frac{8}{10} = 0.8$ (lag) $\boxed{\cos\phi = 0.8}$

(iii) $P = \sqrt{3} V_L I_L \cos\phi = \sqrt{3} \times 400 \times 23.1 \times 0.8 = 12800W$

$\boxed{P = 12800W}$

(iv) $S = \sqrt{3} V_L I_L = \sqrt{3} \times 400 \times 23.1 = 16000 VA$ $\boxed{S = 16000 VA}$



Ques A balanced delta connected load of $(12 + j9)$ ohm is connected to 3-phase 400V supply. Find (i) line current (ii) power factor (iii) power drawn (iv) Reactive volt amp. (v) total volt Ample.

Solⁿ Given values $Z_{ph} = (12 + j9)\Omega = \sqrt{(12)^2 + (9)^2} = \sqrt{225}$

$Z_{ph} = 15$

$V_L = 400V = V_{ph}$

(i) $I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400}{15} = 26.66A$

$I_L = \sqrt{3} \times 26.66 = 46.17A$ $\boxed{I_L = 46.17A}$

(ii) $\cos\phi = \frac{R_{ph}}{Z_{ph}} = \frac{12}{15} = 0.8$ (lag) $\boxed{\cos\phi = 0.8 \text{ (lag)}}$

(iii) $P = \sqrt{3} V_L I_L \cos\phi = \sqrt{3} \times 400 \times 46.17 \times 0.8 = 25590W$

$\boxed{P = 25590W}$

(iv) $S = \sqrt{3} V_L I_L = \sqrt{3} \times 400 \times 46.17$

$\boxed{S = 31987.5 VA}$

(v) $Q = \sqrt{3} V_L I_L \sin\phi = \sqrt{3} \times 400 \times 46.17 \times \sin(36.86) = 19188.08 VAR$

$\boxed{Q = 19188.08 VAR}$

Ques A 3-phase voltage source has a phase voltage of 120V and supplies star connected load having impedance of $(24+j36)\Omega$ per phase. Calculate

(i) line voltage

(ii) line current

(iii) Total 3-phase power supplied to the load.

Sol Given values $V_{ph} = 120V$, $Z_{ph} = (24+j36)\Omega$

$= \sqrt{(24)^2+(36)^2} = 43.26\Omega$

(i) $V_L = \sqrt{3}V_{ph} = \sqrt{3} \times 120 = 207.8V$

$\boxed{V_L = 207.8V}$

(ii) $I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{120}{43.26} = 2.77A$ $\boxed{I_{ph} = 2.77A}$

$I_{ph} = I_L$

(iii) $P = \sqrt{3}V_L I_L \cos\phi$

$\cos\phi = \frac{R_{ph}}{Z_{ph}} = \frac{24}{43.26} = 0.554$

$= \sqrt{3} \times 207.8 \times 2.77 \times 0.554$

$\boxed{P = 552.32W}$

Ques Three identical coils connected in delta across 400V, 50Hz, 3-phase ac supply, take a line current of 17.32A at power factor of 0.8 lagging. Calculate (i) the phase current (ii) the resistance and inductance of each coil (iii) the power drawn by each coil.

Soln Given values $V_L = 400V$, $f = 50Hz$, $I_L = 17.32$, $\cos\phi = 0.8$ (lag)

(i) $I_{ph} = \frac{I_L}{\sqrt{3}} = \frac{17.32}{\sqrt{3}} = 10A$ $\boxed{I_{ph} = 10A}$

(iii) $P = \sqrt{3}V_L I_L \cos\phi = \sqrt{3} \times 400 \times 17.32 \times 0.8 = 9599.71W$

$\boxed{P = 9599.71W}$

(ii)

Ques The similar coils each having a resistance of 8Ω and an inductance of 0.0191H in series in each phase is connected across a 400V three phase, 50Hz supply. Calculate the line current, power input, KVA and KVAR taken by load.

Soln Given values $R = 8\Omega$, $L = 0.0191H$, $V_L = 400V$, $f = 50Hz$

$Z = 8 + 6j$

$= \sqrt{8^2 + 6^2}$

$Z = 10\Omega$

$2\pi f L = 2 \times 3.14 \times 50 \times 0.0191$

$= 5.9974 = 6\Omega$

For star connected

$V_L = 400$ $\quad V_L = \sqrt{3} V_{ph}$

$V_{ph} = \frac{400}{\sqrt{3}} = 231V$ $\quad \boxed{V_{ph} = 231V}$

$I_{ph} = \frac{231}{10} = 23.1A$ $\quad \boxed{I_{ph} = 23.1A}$

$I_{ph} = I_L$

$\cos\phi = \frac{R_{ph}}{Z_{ph}} = \frac{8}{10} = 0.8$

$P = \sqrt{3} V_L I_L \cos\phi = \sqrt{3} \times 400 \times 23.1 \times 0.8 = 12800W$

$\boxed{P = 12800W}$

$S = \sqrt{3} V_L I_L = \sqrt{3} \times 400 \times 23.1 = 16000VA$

$\boxed{S = 16000VA}$

$Q = \sqrt{3} V_L I_L \sin\phi = \sqrt{3} \times 400 \times 23.1 \times \sin(36.86)$

$\boxed{Q = 9600.27 VAR}$

Ques A 3-phase, star connected balanced load is supplied by 400V, 50Hz. The load takes a leading current of $100\sqrt{3}A$ & power 20KWA. Calculate power factor of load and resistance & inductance per phase.

Soln Given values $V_L = 400V$ $\quad f = 50Hz$.

$I_L = 100\sqrt{3} A$ $\quad P = 20KWA$

$P = \sqrt{3} V_L I_L \cos\phi$ $\qquad \cos\phi = \frac{P}{\sqrt{3} V_L I_L}$

$$\cos\phi = \frac{20 \times 10^3}{\sqrt{3} \times 400 \times 100 \times \sqrt{3}}$$

$$= \frac{20 \times 10^3}{12 \times 10^4}$$

$$\boxed{\cos\phi = 0.166}$$

$$Z = \frac{V_{ph}}{I_{ph}} = \frac{\frac{V_L}{\sqrt{3}}}{100\sqrt{3}} = \frac{\frac{400}{\sqrt{3}}}{100\sqrt{3}} = \frac{231}{100\sqrt{3}} = \frac{231}{173.205} = 4.001\Omega$$

$$\simeq 4\Omega$$

$$R = Z\cos\phi \quad , \quad R = 4 \times 0.166 = 0.664\Omega$$

$$\boxed{R = 0.66\Omega}$$

$$X_L = \sqrt{Z^2 - R^2} = \sqrt{(4)^2 - (0.66)^2}$$

$$= \sqrt{16 - 0.4356}$$

$$= \sqrt{15.5644}$$

$$\boxed{X_L = 3.94\Omega}$$

$$X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f} = \frac{3.94}{2 \times 3.14 \times 50} = 0.01254\Omega$$

$$\boxed{L = 0.0125\Omega}$$

UNIT [redacted]

Single phase Transformer

Ques 1 :→ A 30 KVA, 2000/200 V, Single phase, 50Hz transformer has a Primary resistance of 3.5 ohms and reactance 4.5 ohms. The Secondary resistance and reactance are 0.015 ohms and 0.02 Ohms respectively. Find i) Equivalent resistance, reactance and Impedence reffered to the Primary Side and Secondary Side ii) Total Copper losses in the transformer.

ANS :- $R_{1e} = 5\Omega$, $X_{1e} = 6.5\Omega$, $Z_{1e} = 8.2\Omega$, $P_{cu(total)} = 1125$ watt

Ques 2 :→ The ohmics Values of the circuit Parameters of a transformer having a turns ratio of 5, are $R_1 = 0.5\Omega$, $R_2 = 0.021\Omega$, $X_1 = 3.2\Omega$, $X_2 = 0.12\Omega$, $R_c = 350\Omega$ and $X_m = 98\Omega$ refered to primary. Draw the approximate Equivalent circuits of the transformer referred to Secondary. Show the numerical Values of the circuit Parameters

ANS :- $R_{2e} = 0.041\Omega$, $X_{2e} = 0.248\Omega$, $R_c' = 14\Omega$, $X_m' = 3.92\Omega$.

Ques 3 :→ A 250 KVA Single phase transformer has iron loss of 1.8 KW. The full load Copper loss is 2000 watts. Calculate i) Efficiency at full load 0.8 lagging P.f. ii) KVA Supplied at max. efficiency (iii) maximum efficiency at 0.8 lagging P.f.

ANS :- $\%\eta = 98.135\%$, KVA at $\eta_{max} = 237.17$ KVA, $\%\eta_{max} = 98.137\%$

Ques 4 :→ The efficiency of a 400 KVA, Single phase transformer is 98.77% at full load 0.8 P.f. and 99.13% at half load unity P.f. Find (i) Iron losses at full and half full loads (ii) Copper losses at full and half off full loads.

ANS :- $P_i = 1012.0225$ watt. $P_{cu(full\,load)} = 2972.9931$ watt.

P_{cu} on half load $= x^2 P_{cu(F.L.)} = 743.2482$ watt.

The maximum efficiency of a 100 kVA transformer is 98% and operates at 90% full load unity power factor. Calculate the efficiency of a transformer at unity power factor at full load. ANS: $\%\eta_{FL} = 98.3912\%$.

Ques: The maximum efficiency of a 100kVA, 1100/440 Volt 50Hz transformer is 96%. This occurs at 75% of full load at 0.8 pf lagging. Find the efficiency of Transformer at 3/4 F.L. at 0.6 p.f. leading. ANS: $\%\eta = 94.736\%$.

USING FORMULAES

Rated Equivalent circuit :- Referred to Primary Side.

(1) Equivalent Resistance $= R_{1e} = R_1 + R_2'$ where $R_2' = R_2/k^2$

(2) Equivalent Reactance $= X_{1e} = X_1 + X_2'$ where $X_2' = X_2/k^2$

(3) Equivalent Impedance $= Z_{1e} = \sqrt{R_{1e}^2 + X_{1e}^2}$

Referred to Secondary Side :- (i) Equivalent Resistance $R_{2e} = R_2 + R_1'$, $R_1' = k^2 R_1$

(ii) Equivalent Reactance $= X_{2e} = X_2 + X_1'$ where $X_1' = k^2 X_1$

(iii) Equivalent Impedance $= Z_{2e} = \sqrt{R_{2e}^2 + X_{2e}^2}$

(iv) Internal Parameters $R_e' = k^2 R_e$ and $X_m' = k^2 X_m$.

Related to efficiency :- (i) $\%\eta = \dfrac{x V_2 I_2 \cos\phi_2}{x V_2 I_2 \cos\phi_2 + P_i + x^2 P_{cu}}$

(ii) Max. efficiency $\eta_{max} = \dfrac{x \cdot \text{KVA at } \eta_{max} \times \cos\phi}{x \cdot \text{KVA at } \eta_{max} \times \cos\phi + 2P_i}$

(iii) Corresponding load at η_{max} or KVA at η_{max} = Rated KVA $\sqrt{P_i/P_{cu}}$

(v) Copper loss at x-load $= x^2 P_{cu[F.L.]}$.

Permeability

The flow of flux produced by the magnet not only depends on the magnetic field strength but also related to the medium in which magnet is placed. The force exerted by one magnetic pole on other depends on the medium in which magnets are placed.

Key Point The permeability is defined as the ability or ease with which the magnetic material forces the magnetic flux through a given medium.

For any magnetic material, there are two permeabilities,
i) Absolute permeability ii) Relative permeability.

Absolute Permeability (μ)

The magnetic field strength (H) decides the flux density (B) to be produced by the magnet around it, in a given medium.

The ratio of magnetic flux density B in a particular medium (other than vacuum or air) to the magnetic field strength H producing that flux density is called absolute permeability of that medium

It is denoted by μ and mathematically can be expressed as,

$$\boxed{\mu = \frac{B}{H} \quad \text{i.e } B = \mu H}$$

The permeability is measured in units henries peimetre denoted as H/m.

Permeability of free space or Vacuum (μ_0)

If the magnet is placed in a free space or vacuum or in air then the ratio of flux density B and magnetic field strength H is called Permeability of free space or Vaccum or air.

It is denoted as μ_0 and measured in H/m. It denotes the ease with which the magnetic flux permeates the free space or vacuum or air.
It is experimentally found that this μ_0 i.e ratio of B and H in vacuum remains constant every where in the vacuum and its value is $4\pi \times 10^{-7}$ H/m.

$$\boxed{\mu_0 = \frac{B}{H} \text{ in vacuum} = 4\pi \times 10^{-7} \text{ H/m}}$$

Key Point For a magnetic material, the absolute permeability μ is not constant. This is because B and H bears a nonlinear relation in case of magnetic materials.
If magnetic field strength is increased, there is change in flux density B but not exactly proportional to the increase in H.
The ratio B to H is constant only for free space, vacuum or air.

3. Relative Permeability (μ_r)
Generally the permeability of different magnetic materials is defined relative to the permeability of free space (μ_0).
The relative permeability is defined as the ratio of flux density produced in a medium (other than free space) to the flux density produced in free space, under the influence of same magnetic field strength and under identical conditions.
Thus if the magnetic field strength is H which is producing flux density B in the medium while flux density B_0 in free space then the relative permeability is defined as,

$$\boxed{\mu_r = \frac{B}{B_0}}$$ where H is same.

It is dimensionless and has no units.

For free space, vaccum or air, $\mu_r = 1$

According to definition of absolute permeability we can write for given H,

$\mu = \frac{B}{H}$ in medium ....(1) and $\mu_0 = \frac{B_0}{H}$ in freespace ...(2)

Dividing (1) and (2), $\frac{\mu}{\mu_0} = \frac{B}{B_0}$ but $\frac{B}{B_0} = \mu_r$ i.e. $\frac{\mu}{\mu_0} = \mu_r$

$\therefore$ $\boxed{\mu = \mu_0 \mu_r \ H/m}$

The relative permeability of metals like iron, steel varies from 100 to 100,000.

Magnetomotive Force (M.M.F or F)

$$\boxed{m.m.f = NI \quad \text{ampere turns}}$$

Its unit is ampere turns (AT) or amperes (A).

Reluctance (S)

$S \propto \frac{l}{a}$ where 'l' in 'm' while 'a' in $'m^2'$

$\therefore$ $S \propto \frac{Kl}{a}$

Where K = constant = Reciprocal of absolute permeability of material $= \frac{1}{\mu}$

$$\boxed{S = \frac{l}{\mu a} = \frac{l}{\mu_0 \mu_r a} \ A/Wb}$$

It is measured in amperes per weber (A/Wb) or ampere-turns per weber (AT/Wb).

$\therefore$ Reluctance $= \frac{m.m.f}{flux}$ i.e. $\boxed{S = \frac{NI}{\phi} \ AT/Wb \text{ or } A/Wb}$

Permeance

$$\boxed{\text{Permeance} = \frac{1}{\text{Reluctance}}}$$

It is measured in Weber per ampere (Wb/A).

Example 1. A coil of 2000 turns is wound uniformly over a non magnetic ring of mean circumference of 80 cm and cross sectional area of 0.6 sq. cm. If the current through the coil is 2 A, Calculate i) Magnetising force ii) Reluctance iii) Total flux and iv) Flux density

Solution. (i) $H = 5000\ AT/m$

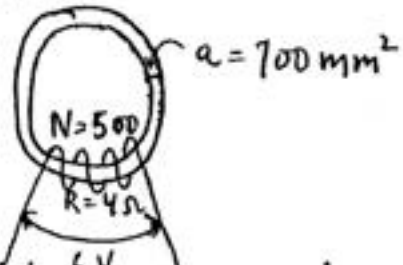
(ii) $S = 1.061 \times 10^{10}\ AT/Wb$

(iii) $\phi = 0.377\ \mu Wb$

(iv) $B = \frac{\phi}{a} = 6.2833 \times 10^{-3}\ Wb/m^2$

Example 2. A coil of insulated wire of 500 turns and of resistance 4 Ω is closely wound on an Iron ring. The ring has a mean diameter 0.25 m and a uniform cross sectional area of 700 mm. Calculate the total flux in the ring when a d.c. supply of 6V is applied to the ends of the winding. Assume a relative permeability of 550.

Solution. $\phi = \frac{m.m.f}{S} = 462\ \mu Wb$



Example 3. A ring of ferromagnetic material has a circular cross-sectio The inner diameter is 7.4 inch, the outer diameter is 9 inch and the thickness is 0.8 inch. There is a coil of 600 turns wound on the ring. When the coil carries a current of 2.5 A, the flux produced in the ring is 1.2×10^{-3} Wb. Find (i) Magnetic field intensity (ii) Reluctance (iii) Permeability.

Example 2. An electromagnet has an air gap of 4mm and flux density in the gap is 1.3 Wb/m². Determine the ampere turns for the gap

Solution. $NI = 4138.028\ AT$

Example 3. An electromagnet shown in fig. has c.s. area of 12cm². Mean length of iron path is 50 cm, length of air gap is 0.4 cm. It is excited by two coils of 400 turns each at 1A. μ_r for material is 1300. Calculate: (i) Reluctance of magnetic circuit. (ii) Reluctance of air gap (iii) Total Reluctance (iv) Total ϕ (v) flux density of air gap.

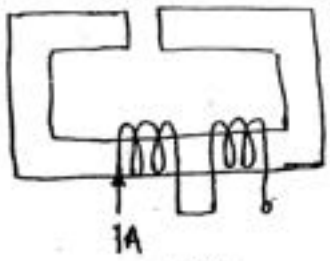
Solution (i) $S_i = 255.056 \times 10^3\ AT/Wb$

(ii) $S_g = 2.65258 \times 10^6\ AT/Wb$

(iii) $S_T = 2.90763 \times 10^6\ AT/Wb$

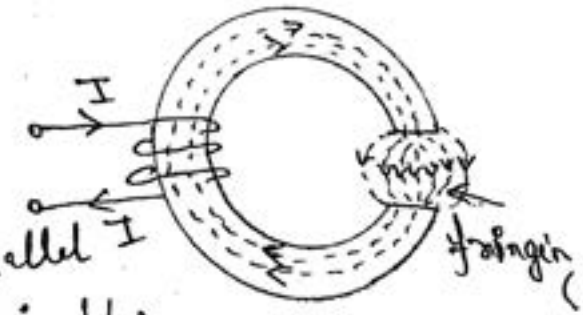
(iv) $\phi_T = 0.27513\ mWb$

(v) $B = 0.2293\ Wb/m^2$



fig

Magnetic fringing



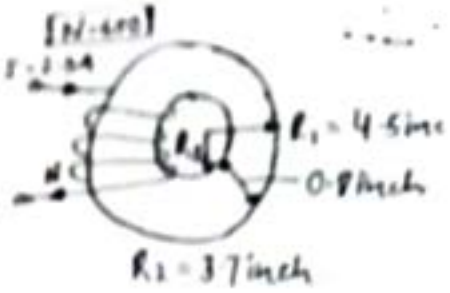
When flux enters into the air gap, it passes through the air gap in terms of parallel flux lines. There exists a force of repulsion b/w the magnetic lines of forces which are parallel and having same direction.

Due to this repulsive force there is tendency of the magnetic flux to bulge out (spread out) at the edge of the air gap. This tendency of flux to bulge out at the edges of the air gap is called magnetic fringing.

Solution (i) H = [illegible] AT/m

(ii) I = $\frac{[illegible]}{\mu_r}$ AT/Wb

(iii) μ_r = [illegible]



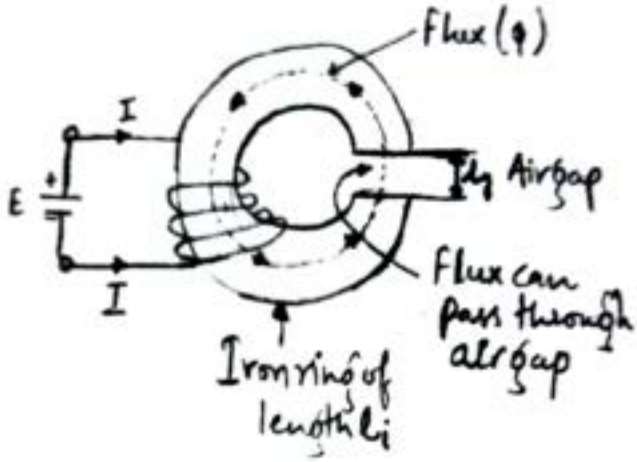
Series Circuit with Air Gap

The series magnetic circuit can also have a short air gap.

Key point This is possible because we have seen earlier that flux can pass through air also.

Such air gap is not possible in case of electric circuit.

$$\boxed{S_i = \frac{l_i}{\mu a_i} \quad \text{and} \quad S_g = \frac{l_g}{\mu_0 a_i}}$$



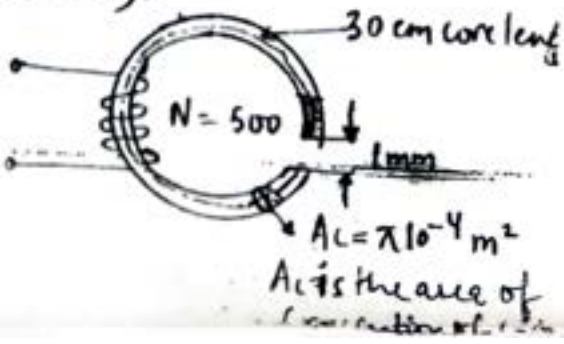
Key point The absolute permeability of air $\mu = \mu_0$ as $\mu_r = 1$ for air.

$$\boxed{NI = S_i\phi + S_g\phi \quad \text{AT for ring}}$$

A ring with an air gap

Example: A wrought iron bar 30 cm long and 2 cm in diameter is bent into a circular shape as given in figure. It is then wound with 500 turns of wire. Calculate the current required to produce a flux of 0.5 mWb in magnetic circuit with an air gap of 1 mm: μ_r (iron) = 4000 (assume constant).

Solution: I = 2.7223 A



It has following two effects:

1. It increases the effective cross-sectional area of the airgap.
2. It reduces the flux density in the airgap.

So leakage, fringing and reluctance, in practice should be as small as possible.

Dot Convention

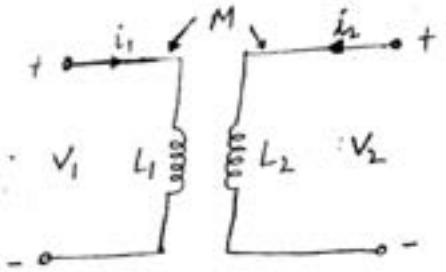
The sign of mutually induced voltage depends on direction of winding of the coils. But it is very inconvenient to supply the information about winding direction of the coils. Hence dot conventions are used for purpose of indicating direction of windin

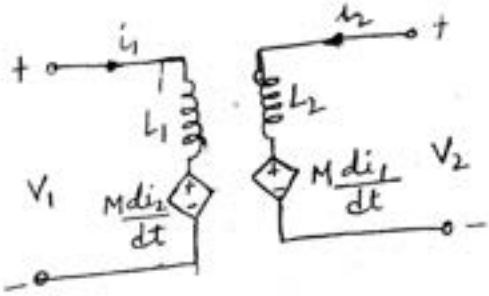
The dot conventions are interpreted as below:

1. If a positive current enters into the dots of both the coils or out of dots of both the coils, then mutually induced voltages for both the coils add to the self induced voltages hence mutually induced voltages will have same polarity as that of self induced voltages.
2. If a positive current enter into (or out of) the dot in one coil and in other coil current flows out of (or into) the dot, then the mutually induced voltages will have polarity opposite to that of self induced voltages.

The dot conventions can be interpreted in simple words as follows:

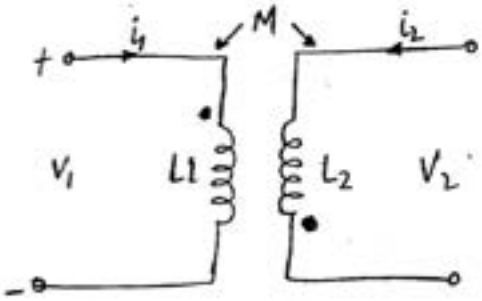
1. If a current enters a dot in one coil, then mutually induced voltage in other coil is positive at the dotted end.
2. If a current leaves a dot in one coil, then mutually induced voltag in other coil is negative at the dotted end.

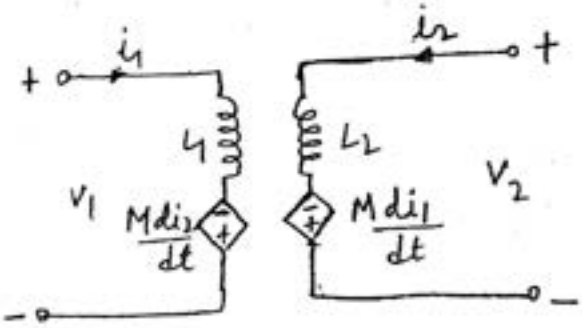




$$V_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$V_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$



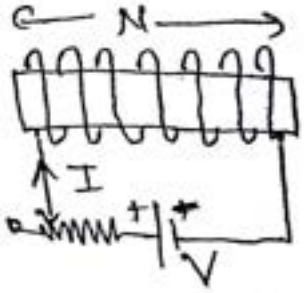


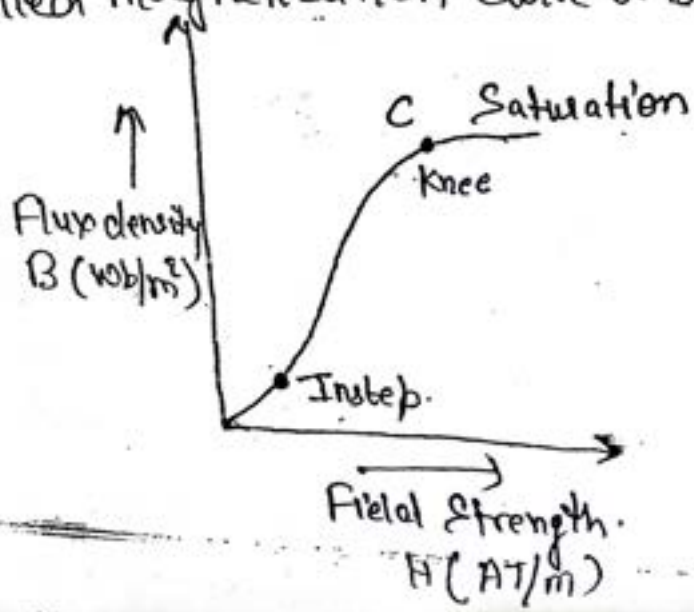
$$V_1 = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

$$V_2 = L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}$$

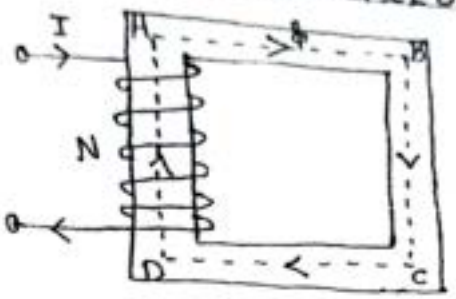
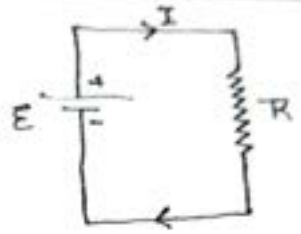
B-H Curve or Magnetization Curve :

The graph bet. the flux density (B) and magnetic field strength (H) for the magnetic material is called Magnetization Curve or B-H-Curve.





Analogy between Electric Circuit And Magnetic Circuit

	Magnetic Circuit SIMILARITIES	Electric Circuit
Circuit		
1- Definition	The closed path followed by magnetic flux is called a magnetic circuit.	The closed path followed by an electric current is called an electric circuit.
2- Driving force	MMF is required to establish flux ϕ in the magnetic circuit and is measured in ampere-turns (AT) or amperes.	EMF is required to cause flow of current in an electric ckt. and is measured in volts.
3- Response	Flux, $\phi = \frac{\text{Driving force}}{\text{Magnetic reluctance}}$ $\phi = \frac{AT}{S}$ webers	Current, $I = \frac{\text{Driving force}}{\text{Electric resistance}}$ $I = \frac{E}{R}$ = amperes
4- Impedance	Reluctance, $S = \frac{1}{\mu_0 \mu_r a} = \frac{1}{\mu a}$ (AT/Wb)	Resistance, $R = \rho \frac{l}{A}$ ohms
(a) For series ckt	$S = S_1 + S_2 + S_3 + \cdots\cdots$ AT/Wb	$R = R_1 + R_2 + R_3 + \cdots\cdots$
(b) For parallel ckt.	$S = \frac{1}{\frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3} + \cdots} \frac{AT}{Wb}$	$R = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \cdots\cdots}$

5- Admittance	Permeance $= \frac{1}{\text{Reluctance}}$ in Wb/AT	Conductance $= \frac{1}{\text{Resistance}}$ in siemens
6- Proportionality constant	Reluctivity $= \frac{1}{\text{Permeability}}$ $= \frac{1}{\mu}$ in m/H	Resistivity $\rho = \frac{1}{\text{Conductivity}}$ in ohm/metre
7- Density	Flux density, $B = \mu \times$ magnetic field intensity $= \mu H$ in tesla or Wb/m^2	Current density, $J = \frac{\text{Electric field intensity}}{\rho}$ in A/m^2
8- Field Intensity	Magnetic field intensity $H = \frac{MMF}{l} = \frac{NI}{l}$ in $\frac{AT}{m}$	Electric field intensity $= \frac{E}{l}$ in volts/meter
9- Drop	MMF Drop $= \phi \times$ reluctance $= \phi S$	Voltage Drop, $= I \times$ Resistance $= IR$